

CHRISNERG MQOBO

Full Stack Software Engineer

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github.com/Chrisnerg | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full Stack Software Engineer with production experience building and shipping enterprise monitoring software deployed across 50+ industrial and energy-sector sites. Strong in React, TypeScript, Node.js, Express.js, and SQL Server (T-SQL), with hands-on work in REST APIs, WebSocket real-time systems, Electron desktop applications, computer vision (Python, OpenCV), and Playwright test automation. Comfortable owning features end to end, from development to on-site deployment and production support.

EXPERIENCE

Junior Software Engineer — Tontrac Systems Analytics

October 2025 - Present

Sandton, Gauteng

- Develop and maintain a real-time monitoring platform (React, TypeScript, Electron, Node.js) used daily by IT Support and Control Room teams across 50+ operational sites, including large-scale energy and industrial-sector deployments.
- Design and consume REST APIs and SQL Server (T-SQL) backed services supporting live production data flows, service integrations, and operational workflows.
- Implement secure IPC communication and WebSocket-driven state synchronization between desktop clients and backend services, enabling real-time device and application updates.
- Built Playwright end-to-end test suites covering critical production workflows, reducing manual regression testing effort by approximately 40%.
- Contribute to computer-vision features in Python and OpenCV, including OCR for automated weighbridge data capture and license-plate recognition that validates vehicle details and strengthens data integrity.
- Engineered Windows OS integrations using Registry APIs and Win32 broadcasts to automate system configuration, eliminating repetitive manual setup on client machines.
- Perform on-site deployments, configuration, and site setup of desktop applications in client production environments, and provide ongoing production support and troubleshooting.
- Partner with engineering, IT Support, and Control Room teams to deliver features, resolve production issues, and lead onboarding and knowledge transfer for new branch operations.

Informatics Tutor — University of Johannesburg

February 2025 - September 2025

Auckland Park, Johannesburg

- Mentored 100+ undergraduate students in programming fundamentals, debugging, and algorithmic thinking, collaborating with academic staff on curriculum workshops.

PROJECTS

URL Shortener — Full Stack Web Application

github.com/Chrisnerg/url-shortener-fullstack

- Built a full stack URL shortening service with click analytics using TypeScript, Node.js, Express.js, and React.
- Designed the REST API with input validation, short-code generation, and persistent link storage.

Web Content Scraper — Automation Tool

github.com/Chrisnerg/Web-Content-Scraper

- Developed a Node.js tool that automates extraction of structured content from web pages.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL (T-SQL), HTML5, CSS3

Frontend: React, Next.js, Tailwind CSS, Responsive UI

Backend: Node.js, Express.js, REST API Development, WebSockets, Authentication, Data Validation

Databases: SQL Server, PostgreSQL, MongoDB, Firebase

Desktop & Systems: Electron, IPC Architecture, Real-Time Systems, Windows Registry / Win32 APIs

Computer Vision: OpenCV, OCR, License-Plate Recognition

Testing & Tools: Playwright, End-to-End Testing, Git, GitHub, Docker, CI/CD, Postman

EDUCATION & CERTIFICATIONS

BSc Computer Science and Informatics — University of Johannesburg

Completed May 2026

Minor in Mathematics and IT Management | Auckland Park, Johannesburg

AWS Certified Developer - Associate | **Microsoft Azure Fundamentals (AZ-900)** — In progress, 2026